

Square-Root Parity Leakage and a Separate-Majorant Barrier

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Abstract

We combine two archived small-noise theorems. Fixed-length Gaussian localization makes the raw trace error $O_n(\sigma^2)$, while the negative parity eigenvalue satisfies $1 + \lambda_-(\sigma) = C_*\sqrt{\sigma} + o(\sqrt{\sigma})$. After Hardy scaling and peripheral extraction, every fixed bulk coefficient therefore has an explicit square-root leading term. More strongly, the scalar parity factor can be expanded uniformly through logarithmic order. Its separate absolute weighted budget grows on the minimal bridge clock because $R/r_H > 1$. Thus a triangle-inequality proof that majorizes raw trace and parity errors separately cannot supply the missing full-trace bridge. This does not prove that their combined, parity-extracted error diverges; moving-order cancellation remains an admissible route.

1 Bulk coefficient law

Let P_n be the deterministic flat trace and let

$$c_{\sigma,n}^H = r_H^{-n} \{\text{Tr } K_\sigma^n - 1 - \lambda_-(\sigma)^n\}, \quad c_n^H = r_H^{-n} \{P_n - 1 - (-1)^n\}, \quad r_H = 0.85.$$

The fixed-length localization theorem gives $\text{Tr } K_\sigma^n = P_n + O_n(\sigma^2)$, while the boundary-layer theorem gives

$$1 + \lambda_-(\sigma) = C_*\sqrt{\sigma} + o(\sqrt{\sigma}), \quad C_* > 0.$$

Theorem 1.1. *For every fixed $n \geq 2$,*

$$\frac{c_{\sigma,n}^H - c_n^H}{\sqrt{\sigma}} \longrightarrow (-1)^n n C_* r_H^{-n}.$$

Proof. Write $\lambda_- = -(1 - \delta_\sigma)$ with $\delta_\sigma = C_*\sqrt{\sigma} + o(\sqrt{\sigma})$. For fixed n ,

$$\lambda_-^n = (-1)^n \{1 - n\delta_\sigma + o(\sqrt{\sigma})\}.$$

Insert this and the $O_n(\sigma^2)$ raw-trace error into the definition of $c_{\sigma,n}^H$. □

2 Logarithmic separate-majorant barrier

Fix $R = 7/5$ and $h_\sigma = \lceil b \log(1/\sigma) \rceil$. At odd pre-alias orders the deterministic pole and finite counterloop moments vanish. Define the separate parity budget

$$B_\sigma^{\text{par}}(R) = \sum_{\substack{3 \leq n < h_\sigma \\ n \text{ odd}}} \frac{|\lambda_-(\sigma)^n - (-1)^n|}{n} \left(\frac{R}{r_H} \right)^n.$$

Theorem 2.1. *If $b \log(R/r_H) > 1/2$, then $B_\sigma^{\text{par}}(R) \rightarrow \infty$. Its logarithmic growth exponent is*

$$b \log(R/r_H) - \frac{1}{2}.$$

At $b = 1/\log(10/7)$ this exponent equals 0.8990081854606016....

Proof. Uniformly for $n \leq h_\sigma$, one has $n\delta_\sigma \rightarrow 0$ and hence

$$|(1 - \delta_\sigma)^n - 1| = n\delta_\sigma\{1 + o(1)\}.$$

After division by n , the odd geometric sum is dominated above and below by constant multiples of $\sqrt{\sigma}(R/r_H)^{h_\sigma}$. Taking logarithms gives the exponent. $\square$

3 Protocol and boundary

The computation uses the archived value $C_* = 0.105258535936908\dots$ to evaluate the exact scalar parity budget for the leading model over five noise scales. The exponent theorem uses only $C_* > 0$.

Remark 3.1. A divergent term in a separated triangle inequality is not a lower bound on the combined error $c_{\sigma,n}^H - s_{k,n} - a_n$. Raw trace and parity corrections can cancel at moving order. RH-298 therefore rules out one proof architecture, not the physical bridge or the operator route globally. Gates A–E remain false/open.